Minor- II

School of Mathematics and Statistics
University of Hyderabad

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Duration: 60 minutes
Maximum Score: 20 points
Instructor: Dharmendra Kumar
Course: Eng Math - II

Instructions: You may use results proven in the lectures; however, answers without justification will receive a score of zero.

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1. Prove / Disprove:
$$\operatorname{Log}\left[(-1+i)^2\right] \;=\; 2\operatorname{Log}(-1+i),$$
where Log (z) is the principal value of $\log(z)$. [5]

2. Show that the function
$$f(z) \;=\; \frac{\operatorname{Log}(z+4)}{z^2+i}$$
is analytic everywhere except at the points $\pm\frac{(1-i)}{\sqrt{2}}$ and on the portion $x \leq -4$ of real axis. [5]

3. Find the principal value of $(-i)^i$. [5]

4. Evaluate the
$$\int_C \frac{z^2}{z+3}\,dz,$$
where C is the unit circle $|z| = 1$ in anti clockwise direction. [5]

All the best !